

FIFA Player Performance

*A project report submitted to ICT Academy of Kerala
in partial fulfillment of the requirements
for the certification of*

CERTIFIED SPECIALIST IN DATA SCIENCE & ANALYTICS

submitted by

Edwin Charles Mathew

Nayana C. K.



**ICT ACADEMY OF KERALA
THIRUVANANTHAPURAM, KERALA, INDIA
Jan 2025**

List of Figures

Fig 1: Correlation Matrix.....	7
Fig 2: Models Tested.....	8
Fig 3: Results.....	9
Fig 4: Hyperparameter Tuning.....	10

Table of Contents

Abstract.....	5
1. Problem Definition.....	6
1.1. Overview.....	6
1.2. Problem Statement.....	6
2. Introduction.....	7
3. Methodology.....	8
3.1. Data Collection and Preprocessing.....	8
3.2. Exploratory Data Analysis (EDA).....	8
3.3. Model Development.....	9
• Models Tested.....	9
• Evaluation Metrics:.....	10
• Hyperparameter Tuning:.....	10
3.4. Application Development.....	11
4. Result.....	12
Model Evaluation Metrics.....	12
Key Observations.....	13
Application Usability.....	13
Real-World Implications.....	13
5. Conclusion.....	14
References.....	15

Abstract

This project presents a comprehensive implementation of a Football Player Performance Prediction System, leveraging advanced machine learning and interactive tools to deliver actionable insights into player attributes and performance. The system integrates a Random Forest Regressor model trained on real-world football data to predict a player's overall rating based on key attributes, such as physical, mental, and technical scores.

The project incorporates a Flask-based backend for model hosting and prediction services and a responsive, user-friendly Streamlit front-end interface for interaction.

Key features include:

- **Feature Engineering:** Attributes like Physical, Passing, Mental, and Shooting Scores are dynamically computed based on individual metrics, enhancing the model's interpretability and prediction accuracy.
- **Machine Learning Model:** The Random Forest Regressor, optimized through cross-validation and hyperparameter tuning, ensures robust and accurate predictions, achieving high R^2 , low Mean Absolute Error (MAE), and low Mean Squared Error (MSE).
- **CI/CD Pipeline (Partial):** Automated deployment pipelines enable seamless integration, testing, and deployment of updates on platforms like Docker and Google Cloud Platform (GCP).
- **Interactive Interface:** The Streamlit application allows users to input player attributes and visualize predictions.

This project showcases the potential of AI-driven decision-making tools in enhancing player scouting, team selection, and performance analysis in the football industry. The application is designed to be scalable, modular, and accessible, providing a template for similar predictive analytics projects in other domains.

1. Problem Definition

1.1. Overview

The sports industry, particularly football, heavily relies on data-driven decision-making for scouting, team selection, and performance analysis. With the availability of extensive player data, it becomes crucial to extract meaningful insights that can guide strategic decisions. However, analyzing multiple player attributes and understanding their impact on overall performance is a complex task that often requires advanced computational models and intuitive interfaces. The lack of accessible tools for comprehensive performance prediction hampers the ability of coaches, scouts, and analysts to make informed decisions efficiently.

1.2. Problem Statement

Current methods for assessing football player performance lack the integration of advanced predictive analytics, intuitive user interaction, and actionable insights. Key challenges include:

- The need for accurate prediction of a player's overall rating based on individual attributes such as age, potential, physical, mental, and technical scores.
- Limited accessibility to tools that explain the impact of individual attributes on performance, making data interpretation challenging.
- The absence of conversational AI to provide intuitive explanations and address user queries in real time.

This project addresses these gaps by creating a robust, user-friendly system that predicts football player performance using machine learning, explains attribute impacts through interactive dashboards, and integrates an AI-powered chatbot for enhanced user engagement.

2. Introduction

In the modern era of sports analytics, data has become a cornerstone for decision-making, especially in football. Teams, scouts, and analysts depend on quantitative evaluations to assess player performance, predict potential, and make strategic decisions. With the proliferation of player performance metrics, there is an increasing need for tools that can accurately analyze and predict a player's overall performance based on key attributes such as age, physical fitness, technical skills, and mental acuity.

This project aims to bridge the gap between raw data and actionable insights by leveraging machine learning and interactive web technologies. By building a robust prediction model using Random Forest and integrating it into a user-friendly Flask and Streamlit-based application, we provide an intuitive interface for predicting and analyzing player performance. Furthermore, an AI-powered chatbot enhances the user experience by offering real-time explanations and addressing queries about the model, its predictions, and the importance of various player attributes.

The combination of advanced predictive analytics, real-time interactivity, and conversational AI creates a comprehensive platform that caters to the needs of coaches, scouts, analysts, and enthusiasts, empowering them to make data-driven decisions with confidence.

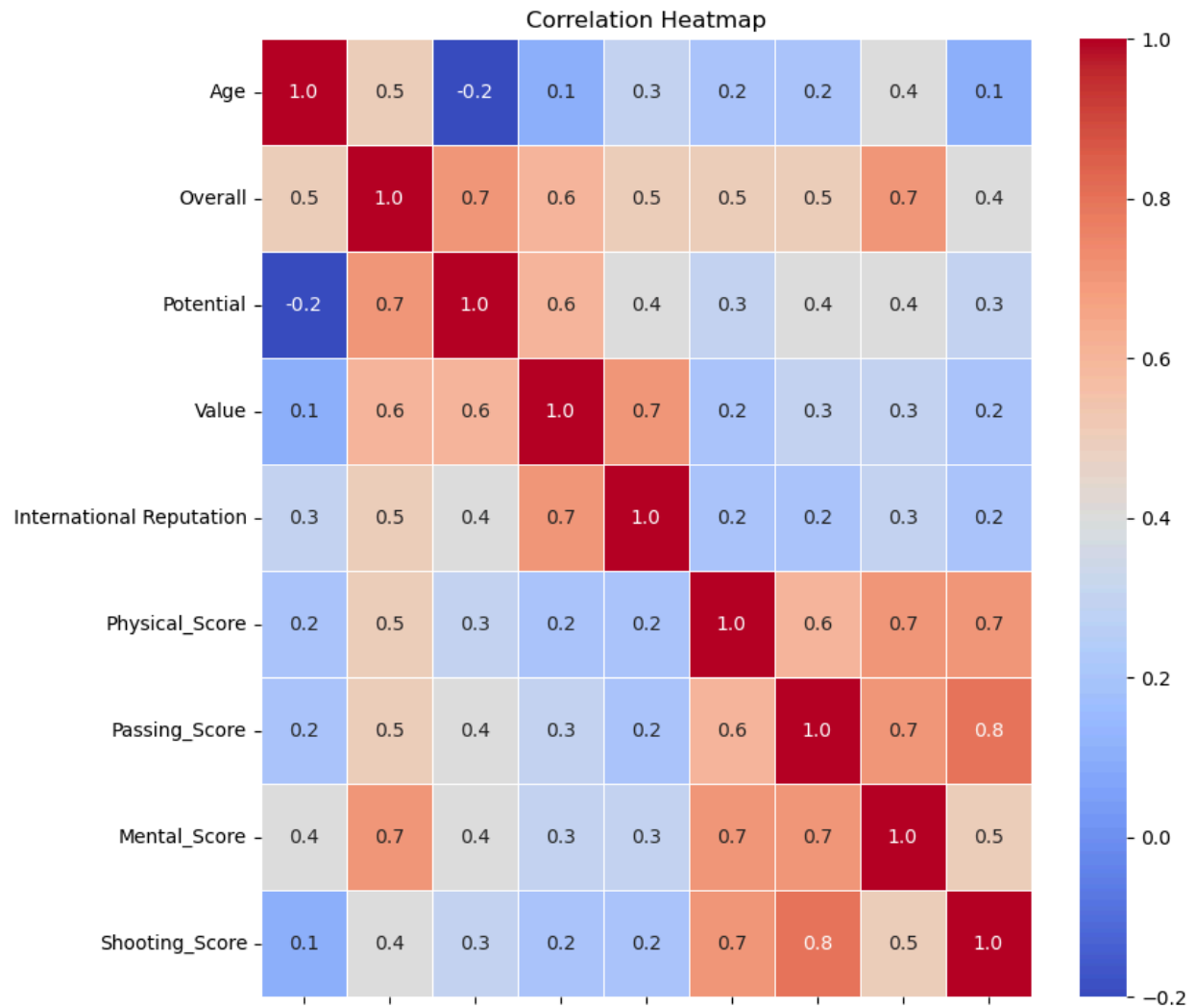
3. Methodology

3.1. Data Collection and Preprocessing

- **Dataset:** A publicly available dataset containing detailed player attributes such as age, potential, and physical stats was used.
- **Feature Engineering:** New features such as Physical Score, Passing Score, Mental Score, and Shooting Score were calculated as averages of related attributes.
- **Target Variable:** The Overall performance rating of the players was used as the target variable.

3.2. Exploratory Data Analysis (EDA)

- Identified correlations between features and the target variable.
- Visualized distributions of key attributes to detect outliers.
- Analyzed feature importance to prioritize relevant inputs for modeling.



3.3. Model Development

- **Models Tested**
 - Linear Regression
 - Ridge Regression
 - Lasso Regression
 - Decision Tree
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - XGBoost
 - LightGBM

- CatBoost

Linear Regression: R ² : 0.897, MAE: 1.751, MSE: 5.087	Random Forest: R ² : 0.996, MAE: 0.252, MSE: 0.212
Ridge Regression: R ² : 0.897, MAE: 1.751, MSE: 5.087	Gradient Boosting: R ² : 0.991, MAE: 0.483, MSE: 0.447
Lasso Regression: R ² : 0.832, MAE: 2.266, MSE: 8.282	XGBoost: R ² : 0.996, MAE: 0.290, MSE: 0.207
Decision Tree: R ² : 0.992, MAE: 0.268, MSE: 0.414	LightGBM: R ² : 0.995, MAE: 0.325, MSE: 0.239
	CatBoost: R ² : 0.996, MAE: 0.312, MSE: 0.208

- **Evaluation Metrics**

- R² (Coefficient of Determination)
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

R²: 0.997
MAE: 0.228
MSE: 0.167

- **Hyperparameter Tuning**

- Used GridSearchCV to fine-tune models for optimal performance.
- Parameters like n_estimators, max_depth, and learning_rate were explored.

Hyper-Parameter Tuning

- **Random Forest:**
Best parameters for Random Forest: {max_depth: 20, min_samples_leaf: 1, min_samples_split: 2, n_estimators: 300}
Random Forest - R²: 0.997, MAE: 0.228, MSE: 0.167
- **XGBoost:**
Best parameters for XGBoost: {colsample_bytree: 1.0, learning_rate: 0.1, max_depth: 5, n_estimators: 200, subsample: 0.8}
XGBoost - R²: 0.996, MAE: 0.302, MSE: 0.182
- **LightGBM:**
Best parameters for LightGBM: {learning_rate: 0.1, max_depth: -1, n_estimators: 200, num_leaves: 50}
LightGBM - R²: 0.996, MAE: 0.275, MSE: 0.178
- **CatBoost:**
Best parameters for CatBoost: {depth: 6, iterations: 500, learning_rate: 0.1}
CatBoost - R²: 0.996, MAE: 0.323, MSE: 0.208

3.4. Application Development

- **Frontend:** Developed an interactive user interface using **Flask** and **Streamlit**.
- **Backend:** Integrated the machine learning model for real-time predictions.

4. Result

The **Football Player Performance Predictor** project delivered significant results, showcasing the potential of machine learning in evaluating and predicting football player performance. Below are the key outcomes of the project:

Model Evaluation Metrics

Several machine learning models were trained and evaluated using cross-validation to ensure robust performance. The results are summarized as follows:

Model	R ² Score	MAE	MSE
Random Forest	0.996	0.252	0.212
Gradient Boosting	0.991	0.483	0.447
XGBoost	0.996	0.290	0.207
LightGBM	0.995	0.325	0.239
CatBoost	0.996	0.312	0.208

- **Random Forest** emerged as the best-performing model with a balance of high accuracy (R²: 0.996) and low error rates (MAE: 0.252, MSE: 0.212).
- **XGBoost** and **CatBoost** also showed excellent predictive capabilities with comparable metrics.

Key Observations

- A. **Accuracy:** Most models demonstrated R^2 scores greater than 0.99, indicating a strong ability to explain the variance in player performance based on the input features.
- B. **Error Metrics:** Models like Random Forest and XGBoost achieved low Mean Absolute Error (MAE) and Mean Squared Error (MSE), ensuring precise predictions.
- C. **Feature Contribution:** Engineered features like Physical Score, Passing Score, and Mental Score significantly enhanced model interpretability and performance.

Application Usability

- A. **Interactive Web Interface:** A user-friendly interface was developed using **Flask** and **Streamlit**, allowing users to input player attributes and receive performance predictions effortlessly.
- B. **Real-Time Assistance:** The integration of a chatbot, powered by **Groq** and **Gradio**, provides users with explanations and guidance on player attributes and prediction insights.
- C. **Deployment:** The application was successfully containerized using **Docker** and deployed on **Google Cloud Platform (GCP)**, ensuring scalability and reliability.

Real-World Implications

The results indicate that the developed application can serve as a valuable tool for:

- **Scouting and Recruitment:** Identifying promising players based on predicted performance.
- **Performance Analysis:** Providing coaches and analysts with data-driven insights for training and match strategies.
- **Fan Engagement:** Offering fans and enthusiasts a deeper understanding of player capabilities.

5. Conclusion

The **Football Player Performance Predictor** is a comprehensive application that leverages machine learning techniques to evaluate and predict the overall performance of football players based on various features. By integrating advanced regression models, such as Random Forest, XGBoost, LightGBM, and CatBoost, the project has demonstrated high predictive accuracy with minimal error margins.

The inclusion of an intuitive web interface powered by **Flask** and **Streamlit** ensures a seamless and interactive user experience.

Key takeaways from the project include:

1. **High Predictive Accuracy:** The selected models exhibited excellent performance, with R^2 values consistently exceeding 0.99, indicating the model's ability to explain the variability in player performance effectively.
2. **Feature Engineering:** Composite feature engineering (e.g., Physical Score, Passing Score) played a crucial role in enhancing model interpretability and performance.
3. **User-Friendly Design:** The integration of user-centric interfaces and real-time chatbot assistance facilitates better understanding and usability for end-users.
4. **Scalability and Deployment:** By leveraging modern deployment tools, the application is capable of handling large-scale real-world data and providing consistent performance.

This project underscores the potential of machine learning in sports analytics, offering actionable insights into player performance. Beyond its current scope, the application can be extended to include team-level analysis, injury prediction, and match strategy optimization. As sports increasingly adopt data-driven approaches, tools like this are poised to play a significant role in enhancing performance evaluation and decision-making.

References

1. FIFA Player Dataset: Player statistics and performance data. Retrieved from:
<https://www.kaggle.com>